
Limit Behavior of Optimal Transport Maps

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Abstract

This paper studies the asymptotic behavior of probability distributions under repeated application of optimal transport maps. Working in the Wasserstein space framework, we consider the sequence of pushforward measures generated by iterating a transport map on a given distribution and show that the limiting distribution is determined by the limit of the transport maps themselves.

In the one-dimensional setting, where optimal transport maps are monotone, the dynamics admits neither chaos nor periodic points. We show that the limiting points are characterized by the locations where the cumulative distribution functions of the source and target distributions coincide.

In higher dimensions the dynamics becomes more involved. Certain symmetric cases reduce to one-dimensional structures, while more general situations require numerical investigation. We perform simulations for transport from uniform distributions on bounded domains to truncated Gaussian targets and compute the associated fixed points numerically.

Finally, we analyze the influence of boundary truncation. We prove a stability result showing that if two target densities are close in $C^{0,\alpha}$ on a bounded domain, then the fixed points of the corresponding transport maps are close. As a consequence, if the Gaussian mass outside the truncation region is sufficiently small, the fixed points of the truncated and non-truncated transports remain close.

Keywords: Optimal transport, Wasserstein space, transport dynamics, fixed points.

1 Setting

Let $S \subset \mathbb{R}^2$ be a bounded closed set and let $\mathcal{B}(S)$ denote the Borel σ -algebra on S . Define

$$\mathcal{W} = \{\mu : \mu \text{ is a probability measure on } (S, \mathcal{B}(S)) \text{ with finite second moment}\}.$$

We assume that there exists an underlying probability space $(\Omega, \mathcal{A}, P)$ of $\mathcal{W}$ -valued random variables that induces a probability measure on the space $\mathcal{W}$ with respect to which moments for random variables taking values in $\mathcal{W}$ can be defined.

For any measurable map $T : S \rightarrow S$ and $\mu \in \mathcal{W}$, let $T_{\#}\mu$ denote the pushforward measure of μ , defined by

$$T_{\#}\mu(B) = \mu(\{x \in S : T(x) \in B\}), \quad B \in \mathcal{B}(S).$$

If μ_1 is absolutely continuous with respect to the Lebesgue measure on S , then the 2-Wasserstein distance on $\mathcal{W}$ can be written using the Monge formulation (Villani, 2003) as

$$d_{\mathcal{W}}(\mu_1, \mu_2) = \inf_{T: T_{\#}\mu_1 = \mu_2} \left\{ \int_S \|T(x) - x\|^2 d\mu_1(x) \right\}^{1/2}. \quad (1)$$

A measurable map $T : S \rightarrow S$ satisfying $T_{\#}\mu_1 = \mu_2$ is called a transport map from μ_1 to μ_2 .

In one dimension, the optimal transport map between μ_1 and μ_2 admits an explicit representation given by the monotone rearrangement

$$T = F_2^{-1} \circ F_1,$$

where F_1 and F_2 denote the cumulative distribution functions of μ_1 and μ_2 , respectively. This explicit representation provides a convenient analytical tool for studying optimal transport in the one-dimensional setting [Villani, 2003].

In higher dimensions, such a closed-form representation is no longer available. Instead, Brenier's theorem states that if μ_1 is absolutely continuous with respect to the Lebesgue measure, then the optimal transport map T from μ_1 to μ_2 exists and can be written as

$$T(x) = \nabla \phi(x),$$

where ϕ is a convex potential function [Brenier, 1991, Villani, 2003].

Consequently, the optimal transport problem can be reformulated as a Monge–Ampère equation satisfied by the convex potential ϕ . Denote that ρ_0 and ρ_1 are the probability density of μ_0 and μ_1 respectively. Then the optimal transport map $T = \nabla \phi$ satisfies the Monge–Ampère equation

$$\det(D^2 \phi(x)) = \frac{\rho_0(x)}{\rho_1(\nabla \phi(x))}, \quad x \in S. \quad (2)$$

By Brenier's theorem[Brenier, 1991, Villani, 2003], the Wasserstein space $(\mathcal{W}, W_2)$ satisfies the following ubiquity assumption. For every pair $\mu_0, \mu_1 \in \mathcal{W}$ with $\mu_0 \ll \mathcal{L}^d$, there exists a unique optimal transport map

$$T_{\mu_0}^{\mu_1} : X \rightarrow X$$

such that

$$(T_{\mu_0}^{\mu_1})_{\#}\mu_0 = \mu_1.$$

Moreover, for every $\omega \in \mathcal{W}$, the pushforward measure

$$(T_{\mu_0}^{\mu_1})_{\#}\omega$$

also belongs to $\mathcal{W}$.

This assumption serves as a foundation for the subsequent analysis, since it ensures that the optimal transport map can be iteratively applied to probability measures, while the resulting measures remain in the Wasserstein space $\mathcal{W}$.

2 Limits of Iterated Transport Maps

We are interested in the limiting distribution of the iterated transport

$$\mu_n = (T^n)_{\#}\mu_0.$$

The following lemma shows that studying the limit of the measures μ_n is equivalent to studying the pointwise limit of the transport maps T^n .

Lemma 1. *Let $\mu \in \mathcal{P}_2^{ac}(X)$. If $T^n(x) \rightarrow T^*(x)$ for every $x \in X$, then*

$$\lim_{n \rightarrow \infty} (T^n)_{\#}\mu = (T^*)_{\#}\mu.$$

2.1 The one-dimensional case

In one dimension, the optimal transport map is given by the monotone rearrangement $T = F_1^{-1} \circ F_0$, and therefore is monotone non-decreasing (Villani, 2003). In the one-dimensional case, the optimal transport map T is monotone non-decreasing.

Iterations of monotone maps on an interval cannot exhibit chaotic behavior and do not admit periodic points of period greater than one. Consequently, for every $x \in X$, the sequence $\{T^n(x)\}$ converges to a fixed point of T (Devaney, 1989).

The fixed point x^* of T satisfies

$$T(x^*) = x^*.$$

Since in one dimension the optimal transport map is $T = F_2^{-1} \circ F_1$, we have

$$F_2^{-1}(F_1(x^*)) = x^*.$$

Applying F_2 to both sides gives

$$F_1(x^*) = F_2(x^*).$$

Therefore, the fixed point corresponds to the point where the two cumulative distribution functions intersect.

Further more,

$$T'(x^*) = \frac{d}{dx} F_2^{-1} \circ F_1|_{x=x^*} = \frac{f_{\mu_1}(x^*)}{f_{\mu_2}(x^*)}.$$

The fixed point x^* is attracting if $|T'(x^*)| < 1$, i.e. $f_1(x^*) < f_2(x^*)$.

And The fixed point x^* is repelling if $|T'(x^*)| > 1$, i.e. $f_1(x^*) > f_2(x^*)$.

Some simple observations can be observed. First, attractive fixed points can only occur in regions where

$$\rho_2(x) > \rho_1(x).$$

Second, since $T : [-1, 1] \rightarrow [-1, 1]$ is continuous, the Brouwer fixed point theorem guarantees the existence of at least one fixed point. Under our assumption that $F_1(-1) < F_2(-1)$, the first intersection of the two cumulative distribution functions occurs in a region where $\rho_2(x) > \rho_1(x)$, and hence this fixed point is attracting.

Third, if two attracting fixed points exist, then there must be at least one repelling fixed point between them, which lies in a region where $\rho_2(x) < \rho_1(x)$.

This observation makes us see more information from the pdf plot.

Now we consider the 1-dim case and consider the optimal transport from Uniform Distribution on $X = [-1, 1]$ to a truncated Gaussian on $[-1, 1]$.

To begin with, we consider a truncated Gaussian distribution with mean zero. We compute the fixed point using two approaches: numerical iteration and direct solution of the fixed-point equation. For the numerical approach, we generate 2000 points from the uniform distribution on $[-1, 1]$ and iteratively apply the transport map $T = (F_2^{\text{tr}})^{-1} \circ F_1$ to these points until convergence, which yields the numerical fixed point. For the theoretical approach, we solve the fixed-point equation directly to obtain the analytical solution. The results are shown in Figure 1(a)(b).

In the mean-zero Gaussian case, the attracting fixed point occurs at $x = 0$, where the cumulative distribution functions of both distributions equal $\frac{1}{2}$. The repelling fixed points occur at the boundaries, where the CDF values are 0 and 1, respectively.

For the truncated Gaussian with $\mu = 0.5$, Figure 1(c)(d) shows that the limiting point is not the mode of the truncated Gaussian. Indeed, if x were located at the mode, the Gaussian CDF would equal $\frac{1}{2}$, while the CDF of the uniform distribution at that location is strictly larger than $\frac{1}{2}$. Consequently, the transport map shifts the point to the right until the two CDF values coincide.

We next consider the case of a mixture of two Gaussian components, which leads to a bimodal target distribution.

3 The two-dimensional case

In higher dimensions the structure of optimal transport maps becomes significantly more complicated. This is mainly because we don't have an explicit expression of $T = F_2^{-1} \circ F_1$. Instead, T is determined by the Monge–Ampère equation 2, which makes the analysis and derivation hard.

Optimal Transport from Uniform $[-1,1]$ to Truncated Gaussian

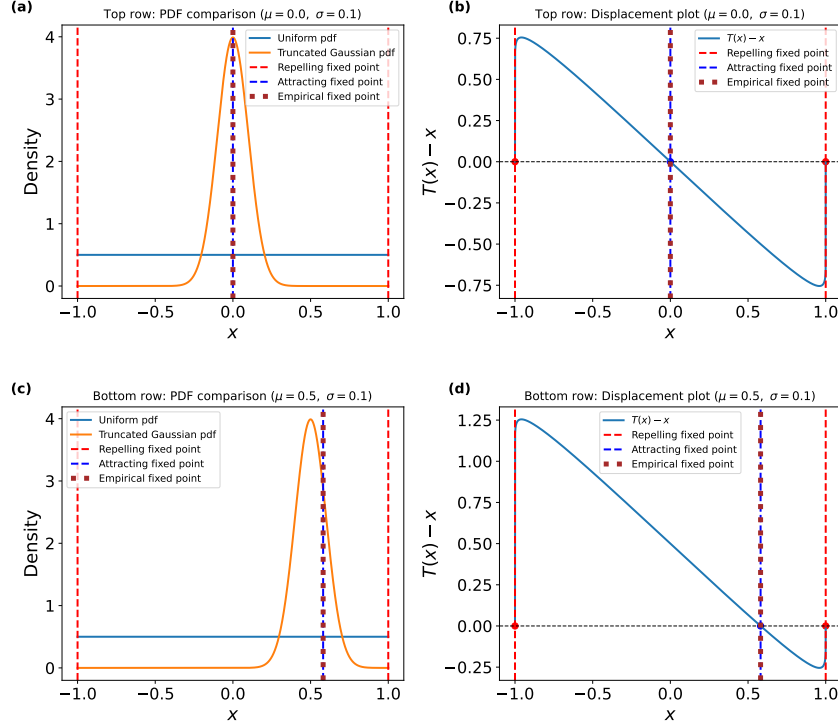


Figure 1: Optimal transport from a uniform distribution on $[-1, 1]$ to truncated Gaussian distributions. (a) PDF comparison for $\mu = 0$ and $\sigma = 0.1$. (b) Displacement function $T(x) - x$ for $\mu = 0$ and $\sigma = 0.1$. (c) PDF comparison for $\mu = 0.5$ and $\sigma = 0.1$. (d) Displacement function $T(x) - x$ for $\mu = 0.5$ and $\sigma = 0.1$.

Simple situations include

1. $T_{\#} \text{Uniform}([-1, 1]^2) \rightarrow N_2(\boldsymbol{\mu}, \Sigma)$, $\Sigma = \text{diag}(\sigma_1^2, \sigma_2^2)$,
2. $T_{\#} : \text{Uniform}(B_1(0)) \rightarrow N_2(\mathbf{0}, \sigma^2 I)$.

These cases can essentially be reduced to one-dimensional problems.

If the Gaussian target distribution has independent components and the source region is $[-1, 1]^2$, then the optimal transport map takes the form

$$T(x_1, x_2) = (T_1(x_1), T_2(x_2)),$$

where each component corresponds to the one-dimensional optimal transport map. To be more specific,

$$T_i = F_{N(\mu_i, \sigma_i^2)}^{-1} \circ F_{\text{Unif}([-1, 1])}.$$

If the source region is a disk centered at the origin, $B_r(0)$, and the covariance matrix of the Gaussian is $\sigma^2 I$, then by rotational symmetry the transport map is radial. In this case the problem reduces to transporting the radial distribution, which is equivalent to a one-dimensional transport problem in the radial coordinate.

For more general situations where the Gaussian distribution is not independent or the source region has a more complex geometry, the optimal transport map T is typically difficult to obtain in closed form. In such cases, one usually resorts to numerical methods. Common approaches include discretizing the Monge–Ampère equation and solving the resulting PDE numerically, or using entropic regularization methods such as the Sinkhorn algorithm to approximate the optimal transport map.

Optimal Transport from Uniform[-1,1] to Truncated Gaussian Mixtures

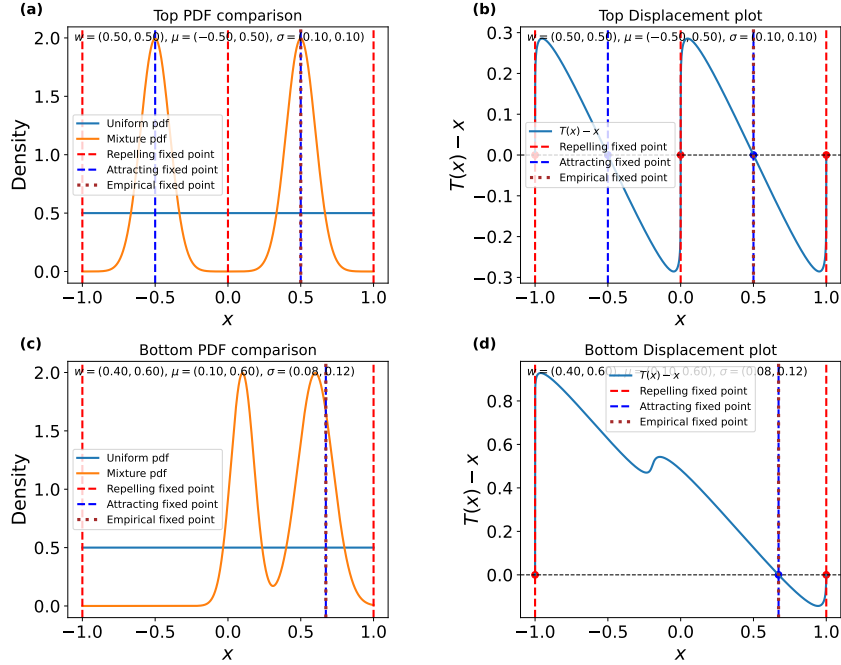


Figure 2: Optimal transport from a uniform distribution on $[-1, 1]$ to truncated Gaussian mixture distributions. (a) PDF comparison for the first mixture setting. (b) Displacement function $T(x) - x$ for the first mixture setting. (c) PDF comparison for the second mixture setting. (d) Displacement function $T(x) - x$ for the second mixture setting.

Here we present a simple simulation for the optimal transport from

$$T_{\#} \text{Uniform}([-1, 1]^2) \rightarrow N_2(\boldsymbol{\mu}, \Sigma).$$

We first generate 2000 points from the uniform distribution on $[-1, 1]^2$. Applying the transport map T iteratively to these points produces simulated limit points of the induced dynamical system. Meanwhile, the theoretical fixed point is obtained by solving the fixed point equation

$$T(x) = x.$$

The comparison between the simulated limit points and the theoretical fixed point is shown in Figure 3.

As illustrated in the figure, the behavior in the independent case agrees with the one-dimensional analysis.

4 Boundary effects

Now we come to the most interesting part in this project, the Boundary effects.

From the analysis above, boundary effects may lead to irregular transport behavior. Here we focus on the following question: when transporting a uniform distribution to a truncated Gaussian distribution, if the Gaussian mass outside the truncation region is very small, are the fixed points of the Gaussian transport map and the truncated Gaussian transport map close?

A related question is how changes of the boundary influence the limiting point. Due to time limitations, we mainly discuss the first question. The main conclusion is that if the Gaussian mass outside the truncation region is sufficiently small, then the fixed points of the Gaussian and truncated Gaussian transport maps are close.

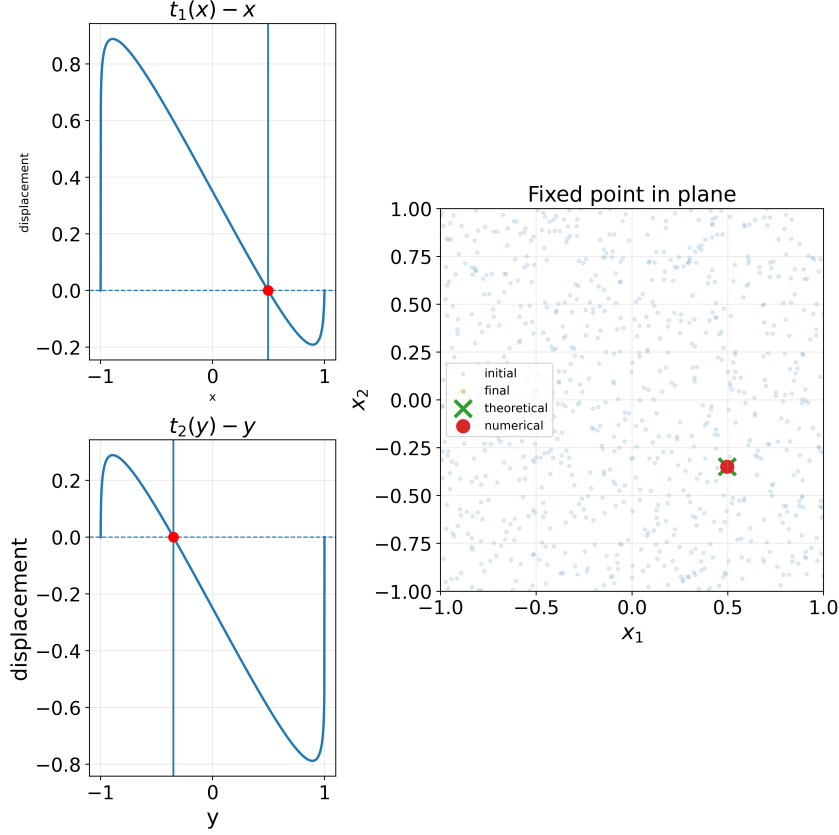


Figure 3: Comparison between simulated convergence points obtained by iterating the transport map T from 2000 samples drawn uniformly from $[-1, 1]^2$ and the theoretical fixed point solving $T(x) = x$.

To justify this, we first establish a more general local stability result. Let μ_0 be supported on a bounded domain, and consider the optimal transport maps $T_1 : \mu_0 \rightarrow \mu_1$ and $T_2 : \mu_0 \rightarrow \mu_2$, with corresponding densities ρ_1 and ρ_2 . Then, in a small neighborhood of a fixed point of T_1 , the difference of the transport maps satisfies

$$\|T_1 - T_2\|_{C^{0,\alpha}} \leq C \|\rho_1 - \rho_2\|_{C^{0,\alpha}},$$

for some constant C depending on the local regularity of the problem.

Consequently, if the Gaussian mass outside the truncation region is small, then $\|\rho_1 - \rho_2\|_{C^{0,\alpha}}$ is small, which implies that T_1 and T_2 are locally close. If a suitable local implicit function argument applies, this further implies that the corresponding fixed points satisfy

$$|x - x^*| \ll 1.$$

The proof proceeds by subtracting the two Monge–Ampère equations and linearizing the resulting equation. Under appropriate regularity assumptions, the linearized equation remains uniformly elliptic. Applying Schauder interior estimates yields local bounds for the difference, and an iteration argument allows us to control the behavior in a small neighborhood of the fixed point.

This result is consistent with intuition: the behavior near the boundary can be complicated, so global estimates are difficult to obtain, while local estimates near a fixed point are more tractable. Since this part of the argument is still preliminary, some steps require further verification.

Theorem 4.1 (Local stability of the Monge–Ampère potential). *Let $\Omega, \Omega^* \subset \mathbb{R}^n$ be bounded domains. Assume that*

$$0 < \lambda \leq \rho_i \leq \Lambda \quad \text{in } \Omega^*, \quad i = 1, 2,$$

and

$$\rho_i \in C^{1,\alpha}(\Omega^*).$$

Let $\rho_0 \in C^{1,\alpha}(\Omega)$ satisfy

$$0 < \lambda \leq \rho_0 \leq \Lambda \quad \text{in } \Omega.$$

Let ϕ_i be convex Alexandrov/classical solutions of

$$\det D^2 \phi_i(x) = \frac{\rho_0(x)}{\rho_i(\nabla \phi_i(x))}, \quad x \in \Omega,$$

with

$$\nabla \phi_i(\Omega) \subset \Omega^*.$$

If

$$\|\rho_1 - \rho_2\|_{C^{0,\alpha}(\Omega^*)} < \varepsilon,$$

then for every fixed point x_1^* of $T_1 = \nabla \phi_1$ there exists $r > 0$ such that

$$\|\nabla \phi_1 - \nabla \phi_2\|_{C^0(B_r(x_1^*))} \leq C \|\rho_1 - \rho_2\|_{C^{0,\alpha}(\Omega^*)}.$$

To apply the local stability theorem to Gaussian truncation, we restrict attention to a bounded neighborhood $\Omega^* \Subset D^\circ$ containing the image of the fixed-point neighborhood under the transport maps. On such a set, the truncated density coincides with a constant rescaling of the original density:

$$\rho_2(y) = \frac{\rho_1(y)}{1 - \varepsilon}, \quad \varepsilon = \int_{D^c} \rho_1(y) dy.$$

Therefore,

$$\|\rho_2 - \rho_1\|_{C^{2,\alpha}(\Omega^*)} \leq C\varepsilon.$$

Since both densities are smooth and remain bounded above and below away from zero on Ω^* , Caffarelli's interior regularity theory yields the required local regularity of the associated Monge–Ampère potentials. The local stability theorem can therefore be applied in this bounded interior region.

Corollary 1 (Local stability of fixed points under Gaussian truncation). *Let $\Omega \subset \mathbb{R}^n$ be a bounded closed convex set with nonempty interior, and let*

$$\rho_0 = \frac{1}{|\Omega|} \mathbf{1}_\Omega$$

be the uniform density on Ω .

Let ρ_G be a Gaussian density on $\mathbb{R}^n$, and define the truncated and renormalized density

$$\rho_{G,\Omega}(y) := \frac{\rho_G(y) \mathbf{1}_\Omega(y)}{\int_\Omega \rho_G(z) dz}.$$

Let

$$T_1 = \nabla \phi_1 : \rho_0 \mapsto \rho_G, \quad T_2 = \nabla \phi_2 : \rho_0 \mapsto \rho_{G,\Omega}$$

be the associated optimal transport maps.

Assume that x_1^ is a fixed point of T_1 and that*

$$I - DT_1(x_1^*)$$

is invertible.

Let

$$\varepsilon := \int_{\Omega^c} \rho_G(y) dy$$

be the Gaussian mass outside Ω .

Then for ε sufficiently small there exists $r > 0$ such that T_2 admits a unique fixed point

$$x_2^* \in B_r(x_1^*),$$

and

$$|x_2^* - x_1^*| \leq C\varepsilon.$$

fig/ot_fixed_points_2x2.pdf

Figure 4: Approximate fixed points of the optimal transport maps from uniform source distributions to the full Gaussian target and the truncated Gaussian target in two dimensions. Panels (a) and (b) correspond to the irregular domain, while panels (c) and (d) correspond to the pentagon. In all panels, the Gaussian mean is $\mu = (0.25, -0.10)$. For panels (a) and (c),

$$\Sigma = \begin{pmatrix} 0.25 & 0.02 \\ 0.02 & 0.18 \end{pmatrix},$$

while for panels (b) and (d),

$$\Sigma = \begin{pmatrix} 0.02 & 0.01 \\ 0.01 & 0.01 \end{pmatrix}.$$

The estimated Gaussian mass outside the source domain is 0.1772 in panel (a), 4.5×10^{-5} in panel (b), 0.2198 in panel (c), and 0 in panel (d). The blue star denotes the attracting fixed point for the full Gaussian target, while the red circle denotes the corresponding convergence point for the truncated Gaussian target.

We also conduct numerical simulations on two types of source domains. The first domain is a regular pentagon, while the second is an irregular star-shaped region generated by a radial perturbation of a circle.

For each domain, we generate 2000 sample points from the uniform distribution on the region. Target samples are drawn either from a Gaussian distribution or from the corresponding truncated Gaussian distribution restricted to the domain.

To approximate the optimal transport map, we compute the entropic regularized optimal transport plan π using the Sinkhorn algorithm. Given the discrete source samples $\{x_i\}_{i=1}^n$ and target samples $\{y_j\}_{j=1}^m$, the optimal plan π_{ij} is obtained numerically. The transport map is then approximated via the barycentric projection

$$T(x_i) = \frac{\sum_{j=1}^m \pi_{ij} y_j}{\sum_{j=1}^m \pi_{ij}}.$$

Using these discrete values, we construct a continuous approximation of the transport map T by interpolation. Starting from an initial point x_0 , we iterate the dynamical system

$$x_{k+1} = T(x_k)$$

to obtain the numerical limit point of the map.

The theoretical fixed point is obtained by solving the fixed point equation

$$T(x) = x,$$

which is computed numerically by minimizing the objective function $\|T(x) - x\|^2$.

From Figure ??, we observe that when the Gaussian distribution has very small probability mass outside the truncation region, the fixed points of the optimal transport maps from the uniform distribution to the Gaussian target and to the truncated Gaussian target are very close.

This indicates that when the truncation only removes a negligible portion of the probability mass, the resulting optimal transport map remains close to the original map, and consequently their fixed points are nearly identical.

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A Appendix

Proofs

Proof of Lemma 1.

Proof. Let

$$\mu_n = (T^n)_\# \mu.$$

Since X is compact, by Theorem A.1, it suffices to prove

$$\mu_n \rightharpoonup (T^*)_\# \mu.$$

Let $f \in C_b(X)$. By the definition of pushforward measure,

$$\int_X f d\mu_n = \int_X f d((T^n)_\# \mu) = \int_X f(T^n(x)) d\mu(x).$$

Since $T^n(x) \rightarrow T^*(x)$ μ -almost surely and f is continuous,

$$f(T^n(x)) \rightarrow f(T^*(x)) \quad \mu\text{-a.s.}$$

Moreover, since f is bounded on X ,

$$|f(T^n(x))| \leq \|f\|_\infty.$$

Therefore, by the Dominated Convergence Theorem,

$$\int_X f(T^n(x)) d\mu(x) \rightarrow \int_X f(T^*(x)) d\mu(x).$$

Hence

$$\int_X f d\mu_n \rightarrow \int_X f d((T^*)_\# \mu),$$

which implies

$$(T^n)_\# \mu \rightharpoonup (T^*)_\# \mu.$$

□

Proof of Theorem 4.1.

Proof. Let ϕ_1 and ϕ_2 be two convex potentials solving the Monge–Ampère equations

$$\det D^2 \phi_i(x) = \frac{\rho_0(x)}{\rho_i(\nabla \phi_i(x))}, \quad i = 1, 2. \quad (3)$$

Define

$$w := \phi_1 - \phi_2.$$

Subtracting the two equations gives

$$\det D^2 \phi_1 - \det D^2 \phi_2 = \frac{\rho_0(x)}{\rho_1(\nabla \phi_1(x))} - \frac{\rho_0(x)}{\rho_2(\nabla \phi_2(x))}.$$

Using the mean value theorem for the determinant,

$$\det D^2 \phi_1 - \det D^2 \phi_2 = \int_0^1 \operatorname{cof}(D^2 \phi_2 + t D^2 w) : D^2 w dt.$$

Define

$$U^{ij}(x) := \int_0^1 \operatorname{cof}_{ij}(D^2 \phi_2 + t D^2 w) dt.$$

Then

$$U^{ij}D_{ij}w = f_1 - f_2. \quad (4)$$

where

$$f_i(x) = \frac{\rho_0(x)}{\rho_i(\nabla\phi_i(x))}.$$

Hence

$$\begin{aligned} f_1 - f_2 &= \rho_0(x) \left(\frac{1}{\rho_1(\nabla\phi_1)} - \frac{1}{\rho_2(\nabla\phi_2)} \right) \\ &= \rho_0(x) \left[\left(\frac{1}{\rho_1(\nabla\phi_1)} - \frac{1}{\rho_1(\nabla\phi_2)} \right) + \left(\frac{1}{\rho_1(\nabla\phi_2)} - \frac{1}{\rho_2(\nabla\phi_2)} \right) \right]. \end{aligned}$$

Denote

$$F_i(y) := \frac{1}{\rho_i(y)}.$$

Then

$$\frac{1}{\rho_1(\nabla\phi_1)} - \frac{1}{\rho_1(\nabla\phi_2)} = F_1(\nabla\phi_1) - F_1(\nabla\phi_2).$$

Applying the mean value theorem gives

$$F_1(\nabla\phi_1) - F_1(\nabla\phi_2) = \nabla F_1(\xi) \cdot (\nabla\phi_1 - \nabla\phi_2)$$

for some ξ between $\nabla\phi_1$ and $\nabla\phi_2$.

Since

$$\nabla\phi_1 - \nabla\phi_2 = \nabla w,$$

we obtain

$$\rho_0(x) \left(\frac{1}{\rho_1(\nabla\phi_1)} - \frac{1}{\rho_1(\nabla\phi_2)} \right) = \rho_0(x) \nabla F_1(\xi) \cdot \nabla w.$$

Define

$$b_i(x) := \rho_0(x) \partial_i F_1(\xi(x)).$$

Then

$$\rho_0(x) \left(\frac{1}{\rho_1(\nabla\phi_1)} - \frac{1}{\rho_1(\nabla\phi_2)} \right) = b_i(x) D_i w.$$

Finally define

$$h_1 - h_2 := \rho_0(x) \left(\frac{1}{\rho_1(\nabla\phi_2)} - \frac{1}{\rho_2(\nabla\phi_2)} \right).$$

Substituting into (4) yields the linearized equation

$$U^{ij}D_{ij}w + b_iD_iw = h_1 - h_2. \quad (5)$$

Since the densities satisfy

$$0 < \lambda \leq \rho_i(x) \leq \Lambda, \quad i = 0, 1, 2, \quad x \in \Omega,$$

the Monge–Ampère equation

$$\det D^2\phi_i(x) = \frac{\rho_0(x)}{\rho_i(\nabla\phi_i(x))}$$

implies

$$\frac{\lambda}{\Lambda} \leq \det D^2\phi_i(x) \leq \frac{\Lambda}{\lambda}.$$

Since ϕ_i are convex, $D^2\phi_i(x)$ are positive definite matrices. By interior regularity theory for the Monge–Ampère equation, there exist positive constants c_0, C_0 such that

$$c_0I \leq D^2\phi_i(x) \leq C_0I, \quad x \in \Omega'.$$

Define

$$M_t(x) := (1-t)D^2\phi_2(x) + tD^2\phi_1(x), \quad t \in [0, 1].$$

Then $M_t(x)$ is also symmetric positive definite and satisfies

$$c_0I \leq M_t(x) \leq C_0I.$$

Since the cofactor matrix satisfies

$$\operatorname{cof}(A) = (\det A)A^{-1},$$

the eigenvalues of $\operatorname{cof}(M_t)$ satisfy

$$c_0^{n-1}I \leq \operatorname{cof}(M_t) \leq C_0^{n-1}I.$$

Define

$$U^{ij}(x) := \int_0^1 \operatorname{cof}_{ij}(M_t(x)) dt.$$

Then U^{ij} satisfies the uniform ellipticity condition

$$c_0^{n-1}|\xi|^2 \leq U^{ij}(x)\xi_i\xi_j \leq C_0^{n-1}|\xi|^2, \quad \forall \xi \in \mathbb{R}^n.$$

Hence the operator $U^{ij}D_{ij}$ is uniformly elliptic.

Next we consider the first order term

$$b_i(x) = \rho_0(x)\partial_i F_1(\xi(x)) = \frac{\rho_0(x)\partial_i \rho_1(\xi(x))}{\rho_1(\xi(x))^2}.$$

Since ρ_0, ρ_1 is bounded away from zero and belongs to $C^{1,\alpha}(\Omega)$ and Ω is compact. we have

$$|b_i(x)| \leq \frac{\Lambda \|\nabla \rho_1\|_{L^\infty(\Omega)}}{\lambda^2}.$$

Therefore the linearized equation takes the form

$$U^{ij}D_{ij}w + b_iD_iw = h_1 - h_2,$$

where U^{ij} is uniformly elliptic and b_i are bounded coefficients.

By the Schauder interior estimate, for any $U \Subset U' \Subset \Omega$,

$$\|w\|_{C^{2,\alpha}(U)} \leq C \left(\|w\|_{C^0(U')} + \|h_1 - h_2\|_{C^{0,\alpha}(U')} \right). \quad (6)$$

Next, note that

$$h_1 - h_2 = \rho_0(x) \frac{\rho_2(\nabla\phi_2) - \rho_1(\nabla\phi_2)}{\rho_1(\nabla\phi_2)\rho_2(\nabla\phi_2)}.$$

Since $\lambda \leq \rho_i \leq \Lambda$, we have

$$\frac{1}{\rho_1(\nabla\phi_2)\rho_2(\nabla\phi_2)} \leq \frac{1}{\lambda^2}.$$

Therefore,

$$\|h_1 - h_2\|_{C^{0,\alpha}(U')} \leq C \|\rho_2(\nabla\phi_2) - \rho_1(\nabla\phi_2)\|_{C^{0,\alpha}(U')}.$$

Since $\phi_2 \in C^2(\Omega)$ and

$$\|D^2\phi_2\|_{L^\infty(\Omega)} \leq C,$$

the gradient $\nabla\phi_2$ is Lipschitz on Ω . Indeed, for any $x, y \in \Omega$ such that the line segment $[x, y] \subset \Omega$,

$$\nabla\phi_2(x) - \nabla\phi_2(y) = \int_0^1 D^2\phi_2(y + t(x - y))(x - y) dt.$$

Hence

$$|\nabla\phi_2(x) - \nabla\phi_2(y)| \leq \int_0^1 \|D^2\phi_2(y + t(x - y))\| |x - y| dt \leq C|x - y|.$$

Applying Lemma 2 with $g = \rho_2 - \rho_1$ and $\psi = \nabla\phi_2$, we obtain

$$\|\rho_2(\nabla\phi_2) - \rho_1(\nabla\phi_2)\|_{C^{0,\alpha}(U')} \leq C \|\rho_2 - \rho_1\|_{C^{0,\alpha}(\Omega^*)}.$$

Consequently,

$$\|h_1 - h_2\|_{C^{0,\alpha}(U')} \leq C \|\rho_2 - \rho_1\|_{C^{0,\alpha}(\Omega^*)}. \quad (7)$$

Substituting (7) into (6), we obtain

$$\|w\|_{C^{2,\alpha}(U)} \leq C \left(\|w\|_{C^0(U')} + \|\rho_2 - \rho_1\|_{C^{0,\alpha}(\Omega^*)} \right).$$

Equivalently,

$$\|\phi_1 - \phi_2\|_{C^{2,\alpha}(U)} \leq C \left(\|\phi_1 - \phi_2\|_{C^0(U')} + \|\rho_2 - \rho_1\|_{C^{0,\alpha}(\Omega^*)} \right).$$

Let x_1^* be a fixed point of the reference transport map $T_1 = \nabla\phi_1$, i.e.

$$T_1(x_1^*) = x_1^*.$$

Since the transport potential is determined only up to an additive constant, we may normalize $w = \phi_1 - \phi_2$ by requiring

$$w(x_1^*) = 0.$$

Applying Lemma 3, we obtain for every $r > 0$ with $B_r(x_1^*) \subset \Omega$,

$$\|w\|_{C^0(B_r(x_1^*))} \leq r \|\nabla w\|_{C^0(B_r(x_1^*))}.$$

Define

$$M(r) := \|\nabla w\|_{C^0(B_r(x_1^*))}.$$

By the Schauder interior estimate, for $B_{2r}(x_1^*) \Subset \Omega$,

$$\|w\|_{C^{2,\alpha}(B_{2r}(x_1^*))} \leq C \left(\|w\|_{C^0(B_{2r}(x_1^*))} + \|h_1 - h_2\|_{C^{0,\alpha}(B_{2r}(x_1^*))} \right).$$

Since $w(x_1^*) = 0$, Lemma 3 yields

$$\|w\|_{C^0(B_{2r}(x_1^*))} \leq 2r \|\nabla w\|_{C^0(B_{2r}(x_1^*))} = 2r M(2r).$$

Therefore,

$$\|w\|_{C^{2,\alpha}(B_r(x_1^*))} \leq C\|h_1 - h_2\|_{C^{0,\alpha}(B_{2r}(x_1^*))} + 2Cr M(2r).$$

Since

$$M(r) = \|\nabla w\|_{C^0(B_r(x_1^*))} \leq \|w\|_{C^{2,\alpha}(B_r(x_1^*))},$$

we obtain

$$M(r) \leq C\|h_1 - h_2\|_{C^{0,\alpha}(B_{2r}(x_1^*))} + 2Cr M(2r).$$

Set

$$A := C\|h_1 - h_2\|_{C^{0,\alpha}(B_{2r}(x_1^*))}.$$

If $r > 0$ is chosen sufficiently small so that

$$2Cr \leq \frac{1}{2},$$

then

$$M(r) \leq A + \frac{1}{2}M(2r).$$

Iterating the inequality $M(r) \leq A + \frac{1}{2}M(2r)$, we obtain

$$M(r) \leq A \sum_{j=0}^{k-1} 2^{-j} + 2^{-k}M(2^k r) \leq 2A + 2^{-k}M(2^k r).$$

Since $B_{2^k r}(x_1^*) \subset \Omega$, we have

$$M(2^k r) \leq \|\nabla w\|_{C^0(\Omega)}.$$

Hence

$$M(r) \leq 2A + 2^{-k}\|\nabla w\|_{C^0(\Omega)}.$$

Given any $\varepsilon > 0$, choose k sufficiently large such that

$$2^{-k}\|\nabla w\|_{C^0(\Omega)} < \frac{\varepsilon}{2}.$$

Then choose $r > 0$ sufficiently small so that

$$2^k r \leq r_0.$$

It follows that

$$M(r) \leq 2A + \varepsilon < C\|h_1 - h_2\|_{C^{0,\alpha}(B_{2r}(x_1^*))} < C\varepsilon.$$

□

Proof of Corollary 1.

Proof. Let

$$\varepsilon := \int_{\Omega^c} \rho_G(y) dy.$$

Then

$$\int_{\Omega} \rho_G(y) dy = 1 - \varepsilon,$$

and hence the truncated and renormalized Gaussian density can be written as

$$\rho_{G,\Omega}(y) = \frac{\rho_G(y)\mathbf{1}_{\Omega}(y)}{1 - \varepsilon}.$$

Since x_1^* is an interior fixed point of T_1 and $T_1 = \nabla\phi_1$ is continuous, there exists $r > 0$ such that

$$\overline{B_r(x_1^*)} \subset \Omega^\circ$$

and $T_1(B_r(x_1^*))$ is bounded. Hence one can choose a bounded open set

$$\Omega^* \Subset \Omega^\circ$$

such that

$$T_1(B_r(x_1^*)) \subset \Omega^*.$$

Since $\Omega^* \Subset \Omega^\circ$, one has

$$\mathbf{1}_\Omega(y) = 1, \quad y \in \Omega^*,$$

and therefore

$$\rho_{G,\Omega}(y) = \frac{\rho_G(y)}{1-\varepsilon}, \quad y \in \Omega^*.$$

Thus, on Ω^* ,

$$\rho_{G,\Omega}(y) - \rho_G(y) = \left(\frac{1}{1-\varepsilon} - 1 \right) \rho_G(y) = \frac{\varepsilon}{1-\varepsilon} \rho_G(y).$$

Since ρ_G is smooth on $\mathbb{R}^n$, it follows that

$$\rho_G, \rho_{G,\Omega} \in C^{2,\alpha}(\Omega^*),$$

and

$$\|\rho_{G,\Omega} - \rho_G\|_{C^{0,\alpha}(\Omega^*)} = \frac{\varepsilon}{1-\varepsilon} \|\rho_G\|_{C^{0,\alpha}(\Omega^*)} \leq C\varepsilon$$

for ε sufficiently small. Likewise,

$$\|\rho_{G,\Omega} - \rho_G\|_{C^{2,\alpha}(\Omega^*)} \leq C\varepsilon.$$

Because ρ_G is strictly positive and smooth, and Ω^* is bounded, there exist constants

$$0 < \lambda \leq \Lambda < \infty$$

such that

$$\lambda \leq \rho_G \leq \Lambda \quad \text{on } \Omega^*.$$

For ε sufficiently small, the same bounds hold for $\rho_{G,\Omega}$ on Ω^* . Therefore both target densities are bounded above and below away from zero and belong to $C^{1,\alpha}(\Omega^*)$, indeed to $C^{2,\alpha}(\Omega^*)$.

By Caffarelli's interior regularity theorem, the corresponding convex transport potentials ϕ_1 and ϕ_2 are $C^{2,\alpha}$ on a neighborhood of x_1^* . Hence the assumptions of Theorem 4.1 are satisfied on this local bounded region, and therefore

$$\|T_1 - T_2\|_{C^0(B_r(x_1^*))} = \|\nabla\phi_1 - \nabla\phi_2\|_{C^0(B_r(x_1^*))} \leq C\|\rho_G - \rho_{G,\Omega}\|_{C^{0,\alpha}(\Omega^*)} \leq C\varepsilon.$$

In particular, after possibly decreasing r , we may assume that for ε sufficiently small,

$$T_2(B_r(x_1^*)) \subset \Omega^*.$$

Now define

$$F_i(x) := x - T_i(x), \quad i = 1, 2.$$

Then fixed points of T_i are exactly the zeros of F_i . Since x_1^* is a fixed point of T_1 ,

$$F_1(x_1^*) = 0.$$

Moreover,

$$DF_1(x) = I - DT_1(x),$$

and by assumption

$$DF_1(x_1^*) = I - DT_1(x_1^*)$$

is invertible. Therefore, by the inverse function theorem, there exists a neighborhood $U \subset B_r(x_1^*)$ of x_1^* and a constant $c > 0$ such that

$$|F_1(x)| \geq c|x - x_1^*|, \quad x \in U.$$

Since

$$\|F_1 - F_2\|_{C^0(B_r(x_1^*))} = \|T_1 - T_2\|_{C^0(B_r(x_1^*))} \leq C\varepsilon,$$

the map F_2 is a small perturbation of F_1 on U . In particular, for ε sufficiently small, F_2 has a unique zero

$$x_2^* \in U \subset B_r(x_1^*).$$

Equivalently,

$$T_2(x_2^*) = x_2^*,$$

so x_2^* is the unique fixed point of T_2 in $B_r(x_1^*)$.

Finally, since $F_2(x_2^*) = 0$,

$$|x_2^* - x_1^*| \leq c^{-1}|F_1(x_2^*)| = c^{-1}|F_1(x_2^*) - F_2(x_2^*)| \leq c^{-1}\|F_1 - F_2\|_{C^0(B_r(x_1^*))} \leq C\varepsilon.$$

Recalling the definition of ε , we conclude that

$$|x_2^* - x_1^*| \leq C \int_{\Omega^c} \rho_G(y) dy.$$

□

Auxiliary Lemma

Theorem A.1 (Santambrogio [2015]). *Let $(\mu_n)_{n \geq 1}$ and μ be probability measures in $\mathcal{P}_p(X)$, $X \subset \mathbb{R}^d$. Then*

$$W_p(\mu_n, \mu) \rightarrow 0 \text{ iff } \mu_n \rightharpoonup \mu$$

This theorem is from C. Villani, *Optimal Transport for Applied Mathematicians: Calculus of Variations, PDEs, and Modeling*, Springer, 2003.

It mainly discusses the convergence of Wasserstein Distance is equivalent to the weak convergence of the distribution.

Theorem A.2 (Caffarelli interior regularity Caffarelli [1992]). *Let u be a convex Alexandrov solution of*

$$\det D^2 u = f \quad \text{in } \Omega.$$

Assume that

$$0 < \lambda \leq f(x) \leq \Lambda \quad \text{in } \Omega.$$

Then

$$u \in C_{\text{loc}}^{1,\alpha}(\Omega)$$

for some $\alpha \in (0, 1)$ depending only on n, λ, Λ .

Moreover, if

$$f \in C^\alpha(\Omega),$$

then

$$u \in C_{\text{loc}}^{2,\alpha}(\Omega).$$

Theorem A.3 (Schauder interior estimate Gilbarg and Trudinger [2001]). *Let Ω be an open subset of $\mathbb{R}^n$, and let $u \in C^{2,\alpha}(\Omega)$ be a bounded solution of the equation*

$$Lu = a^{ij}(x)D_{ij}u + b^i(x)D_i u + c(x)u = f(x) \quad \text{in } \Omega.$$

Assume that $f \in C^\alpha(\Omega)$ and the coefficients satisfy

$$a^{ij}(x)\xi_i\xi_j \geq \lambda|\xi|^2 \quad \text{for all } x \in \Omega, \xi \in \mathbb{R}^n,$$

for some constant $\lambda > 0$, and

$$|a^{ij}|_{C^\alpha(\Omega)} + |b^i|_{C^\alpha(\Omega)} + |c|_{C^\alpha(\Omega)} \leq \Lambda$$

for some constant $\Lambda > 0$.

Then there exists a constant $C = C(n, \alpha, \lambda, \Lambda)$ such that

$$|u|_{2,\alpha;\Omega} \leq C(|u|_{0;\Omega} + |f|_{0,\alpha;\Omega}).$$

Lemma 2 (Hölder composition estimate Gilbarg and Trudinger [2001] or Evans Evans [2010]). *Let $U \subset \mathbb{R}^n$ be bounded and let $V \subset \mathbb{R}^n$. Assume*

$$g \in C^{0,\alpha}(V), \quad \psi \in C^{0,1}(U), \quad \psi(U) \subset V.$$

Then $g \circ \psi \in C^{0,\alpha}(U)$ and

$$\|g \circ \psi\|_{C^{0,\alpha}(U)} \leq C\|g\|_{C^{0,\alpha}(V)},$$

where the constant C depends only on $\|\psi\|_{C^{0,1}(U)}$.

This estimate is standard; see for example Gilbarg and Trudinger [Gilbarg and Trudinger, 2001, Chapter 6] or Evans [Evans, 2010, Appendix C].

Proof. Let $x, y \in U$. Since $g \in C^{0,\alpha}(V)$,

$$|g(\psi(x)) - g(\psi(y))| \leq [g]_{C^{0,\alpha}(V)} |\psi(x) - \psi(y)|^\alpha.$$

Since ψ is Lipschitz on U ,

$$|\psi(x) - \psi(y)| \leq \|\psi\|_{C^{0,1}(U)} |x - y|.$$

Therefore

$$|g(\psi(x)) - g(\psi(y))| \leq [g]_{C^{0,\alpha}(V)} \|\psi\|_{C^{0,1}(U)}^\alpha |x - y|^\alpha.$$

Taking the supremum over $x \neq y$ gives

$$[g \circ \psi]_{C^{0,\alpha}(U)} \leq C [g]_{C^{0,\alpha}(V)}.$$

Moreover,

$$\|g(\psi)\|_{C^0(U)} \leq \|g\|_{C^0(V)}.$$

Combining the two estimates yields

$$\|g \circ \psi\|_{C^{0,\alpha}(U)} \leq C \|g\|_{C^{0,\alpha}(V)}.$$

This completes the proof. □

Lemma 3. Let $x_1^* \in \Omega$ and let $r > 0$ be such that

$$B_r(x_1^*) \subset \Omega.$$

Suppose $w \in C^1(B_r(x_1^*))$ satisfies

$$w(x_1^*) = 0.$$

Then

$$\|w\|_{C^0(B_r(x_1^*))} \leq r \|\nabla w\|_{C^0(B_r(x_1^*))}.$$

Proof. For any $x \in B_r(x_1^*)$, by the fundamental theorem of calculus along the line segment joining x_1^* and x , we have

$$w(x) - w(x_1^*) = \int_0^1 \nabla w(x_1^* + t(x - x_1^*)) \cdot (x - x_1^*) dt.$$

Since $w(x_1^*) = 0$, it follows that

$$w(x) = \int_0^1 \nabla w(x_1^* + t(x - x_1^*)) \cdot (x - x_1^*) dt.$$

Hence

$$|w(x)| \leq |x - x_1^*| \int_0^1 |\nabla w(x_1^* + t(x - x_1^*))| dt \leq |x - x_1^*| \|\nabla w\|_{C^0(B_r(x_1^*))}.$$

Since $x \in B_r(x_1^*)$, we have $|x - x_1^*| \leq r$. Therefore

$$|w(x)| \leq r \|\nabla w\|_{C^0(B_r(x_1^*))}.$$

Taking supremum over $x \in B_r(x_1^*)$ yields

$$\|w\|_{C^0(B_r(x_1^*))} \leq r \|\nabla w\|_{C^0(B_r(x_1^*))}.$$

□